PARTH SARTHI SAXENA

+91-7568815151 | parthsarthisaxena95@gmail.com | GitHub | LinkedIn | LeetCode | Portfolio

PROFESSIONAL SUMMARY

Software engineering student with strong foundations in Python, C++, and backend development. Built full-stack applications with modular system design and secure authentication. Developed high-performance simulation engines processing 130,000+ data points across 10+ configurations. Solved 300+ LeetCode problems in C++. Seeking backend engineering and SDE internship roles.

EDUCATION

JECRC Foundation, Jaipur May 2027

B.Tech – Electronics & Communication Engineering | CGPA: 8.22/10 (~3.4/4.0)

TECHNICAL PROJECTS

RoamCircle — Travel Matching Platform

GitHub: github.com/parthsarthisaxena/RoamAround

- Developed a full-stack travel companion platform end-to-end — from requirements gathering and schema design through to production deployment on Vercel
- Designed and exposed 15+ API endpoints for trip publishing, partner matching, and persistent messaging — with input validation, structured error handling, and consistent status codes
- Constructed a secure authentication layer with token-based sessions, hashed credentials (12 salt rounds), and cookie-based storage — eliminating XSS and CSRF attack vectors
- Modelled collections for users, trips, matches, and threaded conversations with a scalable schema supporting concurrent operations and cross-session state restoration
- Integrated Gemini API for AI-powered itinerary generation with sub-3-second response times — encapsulated as a standalone service for straightforward replacement or extension
- Architected an extensible compatibility scoring foundation for preference-based matching across destination overlap, travel pace, and budget — structured for zero-migration expansion

Quantitative Strategy Backtesting Engine

GitHub: github.com/parthsarthisaxena/nifty-ema-reversion-strategy

- Constructed a modular, event-driven framework ingesting 130,000+ OHLCV records across 5 years of 15-minute market data — configurable execution parameters and pluggable strategy modules
- Engineered a reusable processing pipeline with trade simulation, partial exit handling, and real cost deduction (brokerage, STT, slippage) — validated across 10+ independent strategy configurations
- Assembled an analytics layer computing returns, peak drawdown, win rate, Sharpe ratio, and risk-adjusted metrics — rendered as equity curves and regime-based performance charts
- Separated concerns across four discrete modules — ingestion, signal computation, trade simulation, and reporting — following clean architecture principles for independent testing and reuse
- Chronicled a 4-version iterative process from initial hypothesis through cost-adjusted validation — providing quantified benchmarks at every research stage

GSR Alpha — Statistical Arbitrage System | MCX Futures

GitHub: github.com/parthsarthisaxena/GSR-Alpha

- Delivered an end-to-end research system across 10 years of live market data (2,511 trading sessions) — covering Z-score signal generation, strategy simulation, friction modelling, and holdout validation
- Assembled a comprehensive cost engine deducting brokerage, STT, exchange levies, GST, SEBI charges, slippage, and Indian progressive income tax at marginal slab rates — total round-trip friction ~0.28–0.35%
- Verified statistical robustness at p=0.011 with zero look-ahead contamination and no autocorrelation inflation — applied rolling correlation filters and VIX regime detection to control false signals
- Achieved Sharpe 4.91, net return +170.88%, Max Drawdown -14.50%, Calmar 11.78 — confirmed across 3 independent holdout periods; rejected pairs-trading and Kelly variants after systematic comparison

SKILLS

Languages: Python, C++, JavaScript, SQL
Backend & APIs: Node.js, Express, REST APIs, JWT Authentication, MongoDB, bcrypt, Modular System Design
Data & Libraries: pandas, numpy, scipy, matplotlib, statsmodels, yfinance, Data Pipelines, Simulation Engines
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Git, Clean Code, Debugging
Concepts: Full-Stack Development, Backend Engineering, API Design, Authentication & Security, Statistical Modelling
Problem Solving: 300+ LeetCode problems solved in C++ (Medium/Hard focus) | Active weekly contest participant